

Relational Spatial Alignment for Vision-Language-Action Models

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qm-ipalab.github.io/RIGID

Abstract: Spatial understanding is essential for robots to perceive their surrounding environments and execute actions. Existing approaches typically encode 3D awareness into vision-language-action (VLA) models by directly aligning their representations with pre-trained 3D foundation models. However, we observe that such direct geometric feature alignment essentially exploits 3D foundation priors as a form of low-rank regularization in the feature space, resulting in severe rank collapse of the pre-trained VLA representations. In contrast, we shift the focus from enforcing point-wise equality to aligning pair-wise relationships. We propose a representation-alignment method that enables the injection of 3D spatial awareness without compromising the model’s original expressivity. Extensive experiments on public benchmarks, our new spatial reasoning benchmark, and real-world manipulation scenarios demonstrate that our approach effectively incorporates 3D spatial representations into VLA reasoning, achieving robust performance and strong open-world generalization across diverse manipulation tasks.

Keywords: vision-language-action models, spatial understanding

1 Introduction

Spatial understanding is fundamental to robotic manipulation, enabling robots to interpret their environment from multi-view RGB observations, estimate distances, localize objects, and infer 3D structure either explicitly or implicitly. Traditional approaches [1, 2] typically follow a *reconstruction-before-action* paradigm, in which an explicit 3D scene is first reconstructed and then used for downstream tasks such as path planning and control [3, 4].

Inspired by the success of end-to-end learning in natural language processing [5] and autonomous driving [6, 7], recent attention has shifted toward Vision-Language-Action (VLA) models, a unified end-to-end learning framework for embodied AI. To incorporate spatial awareness, some prior methods [8, 9, 10, 11] follow the reconstruction-before-action paradigm by leveraging explicit spatial representations, such as depth maps or point clouds, as additional inputs to VLA models. However, such approaches often depend on pre-scanned environments or high-quality sensor observations, both of which can be unreliable in real-world scenarios due to sensor noise, reflective surfaces, and occlusions. Consequently, such methods remain limited in constrained environments and struggle to generalize to open-world problems.

To avoid reliance on explicit 3D representations, recent works [12, 13] shift the focus to encode implicit 3D geometric structure within the latent feature space. Building on recent advances in 3D foundation models [14, 15, 16], this approach transfers geometric knowledge into VLAs through feature-level distillation. While these approaches bypass the need for explicit 3D representations, the underlying mechanism of this latent geometric transfer remains underexplored. More importantly,

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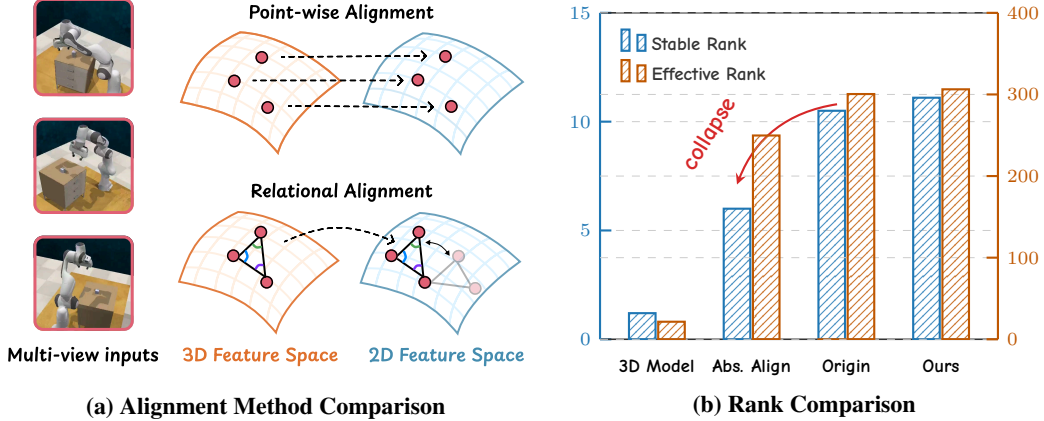


Figure 1: **Spectral and rank analysis of different alignment methods.** (a) Traditional 3D knowledge distillation aligns the point-wise magnitudes (i.e, vector norm) or angles (i.e, cos-similarity) across feature spaces, which often leads to rank collapse (b). In contrast, we align pair-wise relationships to achieve a *rotation-invariant* alignment that effectively prevents feature collapse. See Sec. 3.1 for details. VGGT [16] is used as the 3D teacher model.

it raises a fundamental question: *Does injecting spatial awareness into the VLA with direct feature alignment compromise its original semantic-level representational capabilities?*

In this work, we systematically investigate the direct feature alignment between 3D foundation models and 2D VLA models. Our analysis suggests that direct spatial alignment does not necessarily lead to genuine 3D understanding. Instead, it primarily behaves as a form of low-rank regularization. Specifically, we observe a substantial collapse in the numerical rank of the VLA feature space after alignment (Fig. 1(b); see Sec. 3.1), indicating a significant loss of representation capacity. Consequently, experiments across diverse benchmarks show that models trained with direct spatial alignment suffer from degraded generalization performance.

To overcome these limitations, we propose a new paradigm of relational spatial alignment that aims to distill 3D awareness into 2D VLAs while preserving their expressivity. Motivated by the observation that 3D foundation models encode spatial structures through cosine similarity (i.e., the attention maps) between corresponding features, we argue that effective spatial understanding can be achieved by aligning *pair-wise relationships* rather than enforcing *point-wise equality*. By supervising relative similarities instead of absolute feature directions and norms, our method allows the 2D VLA representations to maintain their native feature coordinate system, with 3D knowledge being encoded in its feature similarity. (Fig. 1(a)). Theoretically, we demonstrate that this relational alignment objective is inherently **rotation-invariant**: The optimization objective remains strictly unchanged under arbitrary orthogonal transformations within the latent feature space, which is crucial for preventing the feature collapse. To practically instantiate this paradigm, we introduce **RIGID: Rotation-Invariant Geometry Distillation**, which implements the relation alignment paradigm and formulates the geometry knowledge transfer as an Optimal Transport problem over spatial attention distributions. Compared to direct feature supervision, RIGID improves spatial awareness while maintaining the numerical rank of the learned representations, effectively preventing capacity collapse. Extensive experiments on both simulated and real-world robotic tasks demonstrate that RIGID achieves state-of-the-art performance, exhibiting better robustness and open-world generalization than direct feature alignment.

2 Related work

Explicit 3D representations in VLA models. Recent advancements seek to enhance the spatial grounding of vision-language-action (VLA) models by directly incorporating geometric informa-

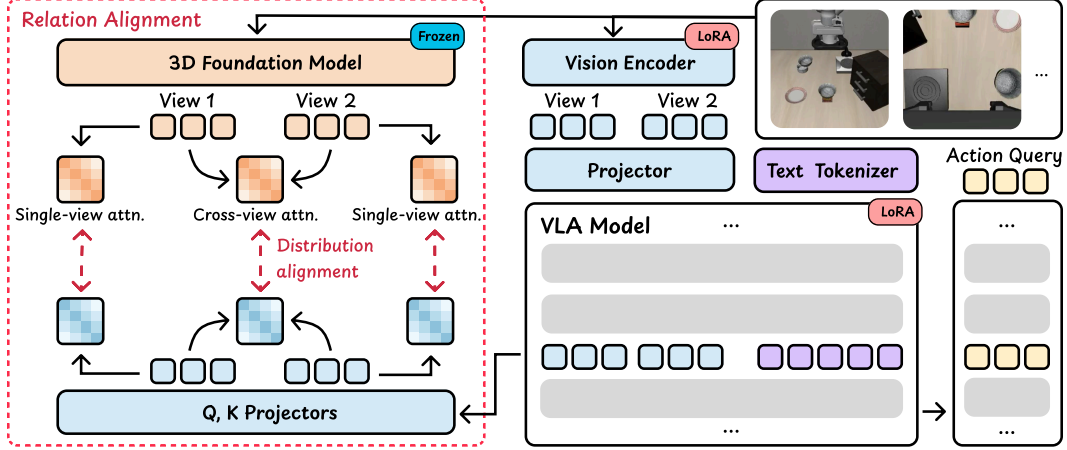


Figure 2: **Overview of our framework.** We extract self- and cross-attention (attn.) maps from both the VLA and the 3D foundation models to serve as relational affinity matrices described in Sec. 3.1. The 3D foundation model is kept frozen, and the VLA model is trained via LoRA. Other parts are fully trainable. More views can be naturally incorporated.

tion. One main stream of methods embeds raw 3D observations, such as depth maps or point clouds, directly into the input space. Frameworks including 3D-VLA [9], 3DS-VLA [17], PointVLA [11], 3D-CAVLA [18], GeoVLA [19], FALCON [20], and BridgeVLA [21] align these 3D inputs with 2D backbone or action experts to unify perception and manipulation, while 4D-VLA [22] extends this by stacking multiple depth maps for temporal context. Another paradigm utilizes auxiliary explicit 3D cues instead of raw sensory data. For instance, SpatialVLA [23] applies Egocentric 3D Position Encoding, and TraceVLA [24] leverages visual traces. Furthermore, approaches like ReconVLA [25], DreamVLA [26], and GuidedVLA [27] enforce 3D representation learning by employing explicit 3D feature reconstruction as an auxiliary supervision signal. Although these models provide precision in handling height variations and object posing, they suffer from substantial data dependencies, as they often require high-fidelity 3D annotations or depth estimations that are frequently noisy in real-world environments.

Implicit 3D representations in VLA models. Recent advances in 3D reconstruction have enabled the end-to-end learning of multi-view geometry [14, 28, 15, 16, 29, 30, 31]. These methods typically employ transformer-based architectures to infer geometric properties such as point maps, depth, and camera parameters from multi-view images. To circumvent the dependency on physical 3D sensors or explicit annotations, recent vision-language-action (VLA) models leverage these learned geometric cues for implicit spatial grounding. Methods such as Evo-0 [32, 33] and GeoAware-VLA [34] directly integrate geometric features generated by 3D foundation models into the VLA backbone. Conversely, techniques including Spatial Forcing [12], ROCKET [13], and GLaD [35] leverage feature distillation. They align the intermediate latent states of the VLA with the geometric embeddings of a frozen teacher model (e.g., VGGT) to acquire spatial awareness. However, this feature integration or direct alignment across different modalities can potentially compromise the 2D image-derived representations acquired during large-scale pre-training, such as semantic understanding and general reasoning abilities, thereby degrading the overall robustness of the model.

Relational knowledge distillation. Relational Knowledge Distillation (RKD) [36] is built upon the knowledge distillation (KD) [37], which focuses on transferring knowledge from the teacher model by distilling the in-batch structure. While traditional RKD methods primarily focus on single-modality and inter-instance associations, our approach targets the latent feature alignment between VLA models and 3D foundation models in spatial through multi-view inputs.

3 Method

3.1 Motivation: feature alignment without representational collapse

Distilling 3D knowledge into 2D VLA with naive feature alignment often leads to representational collapse, as shown in Fig. 1 (b). Specifically, let $\mathbf{Z} \in \mathbb{R}^{M \times C}$ and $\mathbf{T} \in \mathbb{R}^{M \times C_t}$ denote the feature from the student 2D VLA and the frozen 3D teacher model with sequence M and channel C . Existing approaches [12, 13] typically perform point-wise cross-domain alignment:

$$\mathcal{L}_{\text{existing}}(\mathbf{Z}, \mathbf{T}) = - \sum_{i=1}^M \frac{\mathbf{Z}_i \cdot \mathbf{T}_i}{\|\mathbf{Z}_i\| \|\mathbf{T}_i\|}. \quad (1)$$

This objective forces a strict point-wise directional alignment between the two feature spaces. Consequently, the native multimodal semantic structure originally learned by the VLA model is severely distorted, leading to the rank collapse illustrated in Fig. 1.

Instead of enforcing point-wise alignment, we aim to preserve the relational structure between intra-domain features. Specifically, we formulate the objective using the student and teacher Gram matrices, $\mathbf{S}_{\text{vla}} = \mathbf{Z}\mathbf{Z}^\top$ and $\mathbf{S}_{3D} = \mathbf{T}\mathbf{T}^\top$:

$$\mathcal{L}_{\text{RIGID}}(\mathbf{Z}, \mathbf{T}) = \mathcal{D}(\mathbf{Z}\mathbf{Z}^\top, \mathbf{T}\mathbf{T}^\top), \quad (2)$$

where $\mathcal{D}$ is the distance between two matrices. Unlike direct feature supervision, this objective is inherently **rotation-invariant** feature alignment: Given an arbitrary orthogonal transformation $\mathbf{R}$ satisfying $\mathbf{R}\mathbf{R}^\top = \mathbf{I}$, the optimization objective remains unchanged:

$$\mathcal{L}_{\text{RIGID}}(\mathbf{Z}\mathbf{R}, \mathbf{T}) = \mathcal{D}(\mathbf{Z}\mathbf{R}\mathbf{R}^\top\mathbf{Z}^\top, \mathbf{T}\mathbf{T}^\top) = \mathcal{D}(\mathbf{Z}\mathbf{Z}^\top, \mathbf{T}\mathbf{T}^\top) = \mathcal{L}_{\text{RIGID}}(\mathbf{Z}, \mathbf{T}). \quad (3)$$

This rotation invariance allows the VLA model to rotate flexibly within its own semantic coordinate basis, thereby maintaining its original semantic and action representations while still aligning with the geometric structure encoded by the 3D teacher. Since 3D foundation models primarily capture geometry through feature similarity [38], aligning relational affinity effectively transfers spatial awareness without directly constraining original representations.

3.2 RIGID: Rotation-invariant geometry distillation

Given multi-view RGB observations $\{\mathbf{I}_t^{(v)}\}_{v=1}^N$, our goal is to leverage a 3D foundation model as a teacher, and distill its geometry prior into the student VLA model. In modern Transformers, pairwise similarity $\mathbf{Z}\mathbf{Z}^\top$ is naturally encoded in the attention responses with extra non-linearity (see Eq. 4). Therefore, we implement $\mathcal{L}_{\text{RIGID}}$ by distilling the spatial relationships encoded within the 3D teacher model’s self-attention and cross-attention maps. Specifically, let $\mathbf{Z}^{(v)} \in \mathbb{R}^{M \times C}$ denote the visual tokens from the VLA for the v -th view. For any view pair (a, b) , we compute the attention map of the student VLA model using learnable query and key projections $g_q(\cdot)$ and $g_k(\cdot)$:

$$\mathbf{A}_{\text{vla}}^{(a,b)} = \text{Softmax} \left(\frac{g_q(\mathbf{Z}^{(a)})g_k(\mathbf{Z}^{(b)})^\top}{\sqrt{d}} \right), \quad (4)$$

The teacher attention map, $\mathbf{A}_{3D}^{(a,b)}$, is extracted directly from the frozen 3D teacher and used as a distillation target. This attention contains two complementary geometric information: **single-view attention** ($a = b$), which captures intra-view spatial layouts and local geometry, and **cross-view attention** ($a \neq b$), which models multi-view correspondences. The attention from the teacher captures rich geometric cues, providing robust spatial supervision to distill 3D awareness into the student VLA model.

Geometry-aware distribution alignment. Existing distillation objectives, such as the Kullback-Leibler (KL) divergence, enforce strict token-wise correspondence. Under a KL objective, a minor spatial shift is penalized equally to a completely incorrect prediction, offering poor gradient signals when attention distributions do not perfectly overlap. Instead, we formulate $\mathcal{D}$ as an Optimal Transport (OT) problem, optimizing the discrepancy via the Wasserstein distance.

Given the student attention response $\mathbf{A}_{\text{vla}}^{(a,b)}$ and the corresponding teacher attention response $\mathbf{A}_{3D}^{(a,b)}$, we interpret each row as a discrete spatial distribution over image patches. To measure geometric discrepancy between spatial distributions, we define a 2D transport cost matrix $\mathbf{C} \in \mathbb{R}^{M \times M}$ based on patch coordinates on the image grid. Each entry $\mathbf{C}_{p,q}$ represents the Euclidean distance between the spatial locations of the p -th and q -th patches:

$$\mathbf{C}_{p,q} = \|\mathbf{c}_p - \mathbf{c}_q\|_2, \quad (5)$$

where $\mathbf{c}_p$ and $\mathbf{c}_q$ denote the corresponding grid coordinates. We then compute the Wasserstein-based transport distance between the student and teacher attention distributions using entropic-regularized Optimal Transport:

$$\mathcal{L}_{\text{RIGID}} = \mathcal{D}(\mathbf{A}_{\text{vla}}, \mathbf{A}_{3D}) = \frac{1}{V^2 M} \sum_{a=1}^V \sum_{b=1}^V \sum_{i=1}^M \text{Sinkhorn} \left(\mathbf{A}_{\text{vla},i}^{(a,b)}, \mathbf{A}_{3D,i}^{(a,b)}, \mathbf{C} \right), \quad (6)$$

where V denotes the number of views and M denotes total number of query tokens, $\mathbf{A}_{\text{vla},i}^{(a,b)}$ and $\mathbf{A}_{3D,i}^{(a,b)}$ represent the attention distributions of the student and teacher models for the i -th token, respectively, and $\text{Sinkhorn}(\cdot)$ computes the regularized Wasserstein distance.

The final training objective combines the action learning objective and the cross-domain geometric distillation objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{action}} + \beta \mathcal{L}_{\text{RIGID}}, \quad (7)$$

where β controls the strength of geometric supervision. By optimizing the Wasserstein distance between attention distributions, the VLA branch learns geometry-aware spatial reasoning from the 3D teacher while remaining compatible with the original transformer representation space.

4 Simulation

4.1 Setup

Datasets and benchmarks. We train our model on the LIBERO [39] training set. We evaluate our proposed method across three simulated benchmarks: LIBERO, LIBERO-Plus, and our newly introduced LIBERO-Probing suite. We did not utilize the LIBERO-Plus training set.

LIBERO-Probing. We introduce LIBERO-Probing, a new benchmark to systematically evaluate the representation capacity. Specifically, we re-render the LIBERO training environments to obtain ground-truth depth, semantic masks, and spatial correspondences. By freezing the pre-trained VLA and training a lightweight MLP, we can perform an evaluation of the learned representations using the same demonstration states as in the main task.

Implementation details. Our training procedure on LIBERO follows OpenVLA and Spatial Forcing [40, 41, 12]. The base model utilizes a VLM backbone pre-trained on the Open-X-Embodiment dataset [42], incorporating a fused vision encoder that combines features from SigLIP [43] and DINOv2 [44]. To ensure a fair comparison with existing baselines, we train our method using LoRA [45] for [150k] iterations. More details can be found in the supplementary material.

4.2 Benchmark results

In this section, we compare RIGID with state-of-the-art VLAs on simulation benchmarks on different aspects. Detailed analysis of each component, the design choices of VGGT and VLA layers, and other hyperparameters can be found in the supplementary material.

We present a comprehensive evaluation on the LIBERO and the more challenging LIBERO-Plus benchmark. We categorize the baselines into three paradigms: (i) *2D VLAs* trained on purely 2D data, (ii) *Explicit 3D VLAs* requiring additional geometric modalities (e.g., depth or point clouds), and (iii) *Implicit 3D VLAs* that, like ours, distill geometric knowledge in latent space, and thus do not require additional sensors at test time.

Method	Spatial	Object	Goal	Long	Avg
<i>2D VLA</i>					
Diffusion Policy [46]	78.3	92.5	68.3	50.5	72.4
TraceVLA [24]	84.6	85.2	75.1	54.1	74.8
Octo [47]	78.9	85.7	84.6	51.1	75.1
Openvla [40]	84.7	88.4	79.2	53.7	76.5
Dita [48]	84.2	96.3	85.4	63.8	82.4
CoT-VLA [49]	87.5	91.6	87.6	69.0	83.9
π_0 -FAST [50]	96.4	96.8	88.6	60.2	85.5
π_0 [51]	96.8	98.8	95.8	85.2	94.2
UniVLA [52]	96.5	96.8	95.6	92.0	95.2
$\pi_{0.5}$ [53]	98.8	98.2	98.0	92.4	96.9
Openvla-OFT [41]	97.6	98.4	97.9	94.5	97.1
<i>Explicit 3D VLA: w/ depth sensors</i>					
SpatialVLA [23]	88.2	89.9	78.6	55.5	78.1
4D-VLA [54]	88.9	95.2	90.9	79.1	88.6
GeoVLA [19]	98.4	99.0	96.6	96.6	97.7
3D-CAVLA [18]	98.2	99.8	98.2	96.1	98.1
<i>Implicit 3D VLA: w/ RGB inputs only</i>					
GLaD [35]	95.2	95.4	91.9	87.5	92.5
DreamVLA [26]	97.5	94.0	89.5	89.5	92.6
Spatial Forcing [12]	99.4	99.6	98.8	96.0	98.5
Ours	99.6	99.8	98.4	96.1	98.5

Table 1: **Results on LIBERO.**

Model	Cam	Ro	Lang	Light	Back	Noise	Lay	Avg
<i>LIBERO-Spatial</i>								
OpenVLA	88.3	40.0	80.5	98.3	97.3	96.3	93.9	84.0
Spatial forcing	70.2	43.4	87.1	99.3	97.7	88.9	95.1	82.3
Ours	90.4	51.1	77.4	98.6	99.2	96.0	97.9	86.6
<i>LIBERO-Object</i>								
OpenVLA	38.9	25.4	99.0	73.7	97.6	72.3	78.1	66.5
Spatial forcing	49.0	37.9	99.7	99.3	98.0	42.2	72.7	67.7
Ours	49.0	37.2	98.3	97.6	100.0	82.5	73.2	74.3
<i>LIBERO-Goal</i>								
OpenVLA	62.0	25.2	53.2	93.9	92.5	75.2	59.1	63.0
Spatial forcing	74.0	31.8	64.0	95.3	85.4	80.5	64.0	68.6
Ours	74.6	50.6	57.8	90.7	88.6	82.6	67.3	71.1
<i>LIBERO-Long</i>								
OpenVLA	38.7	38.2	87.0	89.4	86.8	63.5	76.9	66.4
Spatial forcing	34.1	35.9	89.9	78.5	83.7	55.0	84.3	63.2
Ours	34.4	50.1	87.5	92.7	88.7	55.0	88.1	67.8
<i>Total four tasks</i>								
OpenVLA-OFT	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
π_0	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
Spatial forcing	56.5	37.0	84.6	93.3	90.8	65.1	78.3	70.3
Ours	62.1	47.3	80.3	94.9	94.1	79.0	81.6	75.0

Table 2: **Results on LIBERO-Plus.**

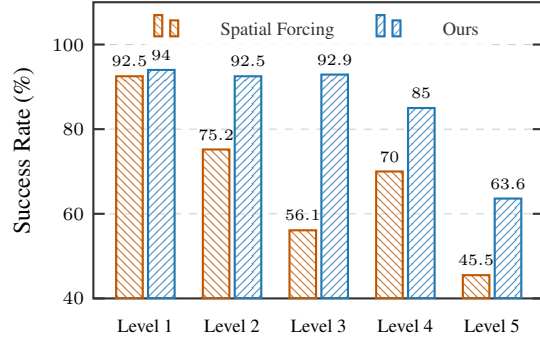
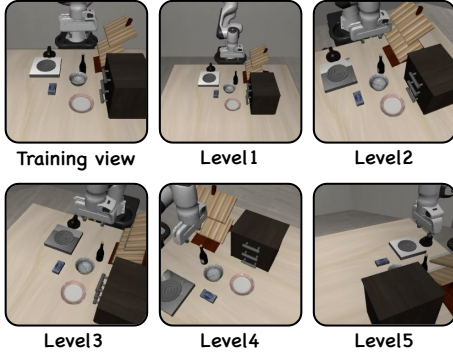


Figure 3: **Robustness to camera view changes.** We evaluate results on camera view change by level on LIBERO-Spatial. Our method outperforms the Spatial Forcing across most levels, showing particular robustness at higher difficulty tiers, mitigating the performance degradation.

On LIBERO, our model consistently matches or outperforms the performance of state-of-the-art implicit and explicit 3D VLAs, demonstrating that our relational alignment successfully distills geometric priors from 3D foundation models. On LIBERO plus (Tab. 2), while naive spatial alignment methods like Spatial Forcing [12] suffer from performance degradation under severe domain shifts due to representational collapse, our method yields superior overall robustness across diverse environmental conditions. This empirical result shows the advantage of aligning relational affinities rather than absolute feature coordinates.

4.3 Representation Evaluation

While downstream task success demonstrates the effectiveness of the learned policy, it does not directly reveal whether the learned representations acquire stronger spatial and semantic understanding. We therefore evaluate the learned representations from two complementary perspectives: robustness to viewpoint changes and explicit semantic/geometric probing.

Task	Metric	OpenVLA -OFT	SF	Ours
Seg.	Pixel Acc. ($\uparrow$)	94.76	96.82	96.94
Depth	$\delta_{1.25}$ ($\uparrow$)	88.12	92.27	93.60
	$\delta_{1.03}$ ($\uparrow$)	25.30	36.88	46.90
	REL (%) ($\downarrow$)	10.93	8.05	6.81
Corr.	MPE ($\downarrow$)	34.10	25.60	30.55
	PCK10 ($\uparrow$)	0.26	0.29	0.33

Figure 4: **Proposed LIBERO-Probing benchmark.** Our benchmark generates the semantic segmentation (Seg.), depth estimation (Depth), and correspondence matching (Corr.) results in the LIBERO training environments to analyze the VLA capabilities.

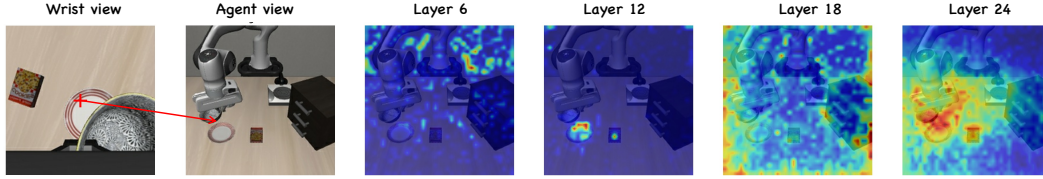


Figure 5: **Cross-view attention across VGGT layers.** For the query point marked in the wrist view, attention maps are overlaid on the agent view. Layer 12 yields a localized response near the corresponding point.

Robustness to viewpoint changes. To evaluate spatial understanding, we further measure robustness to viewpoint variations on LIBERO-Plus. Since these novel viewpoints are excluded from the training set and no viewpoint augmentation is applied during training, this task provides a direct evaluation of the model’s 3D awareness. As shown in Fig. 3, our method consistently outperforms Spatial Forcing across all levels of viewpoint perturbation and maintains an 85% success rate even under the most extreme viewpoint shift (Level 5). These results demonstrate that our rotation-invariant relational alignment effectively distills robust spatial awareness into the VLA representations.

Semantic and geometric probing. We further analyze the semantic and geometric awareness of the learned representations through a set of probing experiments. Specifically, we freeze the learned features and train lightweight MLP heads for downstream probing tasks, including depth estimation, semantic segmentation, and keypoint matching. As in Fig. 4, our method achieves better performance across both semantic and geometric tasks. These results empirically verify that the proposed relational alignment enhances both geometric awareness and semantic understanding compared to direct feature alignment.

Rank intervention. We also examine whether rank preservation contributes to policy performance. We intervene on spectral entropy and find that encouraging higher entropy raises effective rank and success rate, while encouraging lower entropy reduces both for RIGID. These findings support a beneficial role for representation diversity [55], though other optimization effects cannot be excluded. Experimental settings and numerical results are provided in the appendix.

4.4 Component analysis

We examine the distillation targets, view alignment, and teacher layer selection in RIGID. All variants are trained on the LIBERO training set for 30k steps and evaluated zero-shot on the LIBERO-Plus Spatial task suite over 2,402 episodes. Additional ablations of loss functions and other components are provided in the appendix.

Distillation targets. We compare two forms of relational supervision from VGGT: post-softmax query-key attention maps and Gram matrices constructed from hidden features. As shown in

Variant	Cam	Ro	Lang	Light	Back	Noise	Lay	Avg
Gram	68.6	39.3	96.9	100.0	96.9	90.0	95.6	83.9
Direct feature	57.2	42.0	93.3	99.7	100.0	70.1	93.0	78.2
Single-view only	85.9	49.1	75.9	100.0	99.6	93.7	96.1	84.9
Cross-view only	79.0	50.3	79.5	99.0	99.2	92.3	97.7	84.4
RIGID	90.4	51.1	77.4	98.6	99.2	96.0	97.9	86.6

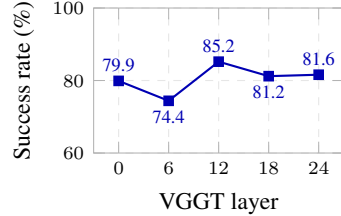


Figure 6: **Component analysis.** (a) Success rates (%) on LIBERO-Plus Spatial; higher is better. Bold indicates the best result. (b) Success rates across VGGT layers. Layer 0 means the output from the VGGT encoder.

Fig. 6(a), the Gram-matrix target achieves comparable or better performance in some perturbation categories but yields a lower success rate under camera perturbations than the attention-based target (68.6% vs. 83.2%). This comparison supports our choice of attention maps for viewpoint robustness. We hypothesize that VGGT’s query and key projections filter out information irrelevant to geometric correspondence, making the resulting attention maps more effective supervision targets. We also examine the contributions of single-view and cross-view alignment in Fig. 6(a). Combining both yields an average success rate of 86.6%, compared with 84.9% for single-view alignment alone and 84.4% for cross-view alignment alone, suggesting that the two provide complementary supervision.

Teacher attention and layer selection. The frozen teacher’s cross-view attention provides the spatial correspondence signal that RIGID aims to distill. Figure 5 illustrates this signal for a query point in the wrist view: layer 12 concentrates attention near the corresponding region in the agent view, whereas layer 6 produces scattered responses and layers 18 and 24 show more diffuse patterns. This example is consistent with the observations of Bratulić et al. [56] on correspondence matching in VGGT’s middle layers. We complement this illustration with a quantitative comparison of attention targets from different teacher layers (Fig. 6b). Layer 12 achieves the highest success rate among the tested layers (85.2%) and is therefore used in RIGID.

5 Real-world experiments

5.1 Experimental setting

Setup. We perform the real robot experiments in a complex toy kitchen environment that contains various objects. In this setting, we define six tasks as shown in Fig. 7: (a) *placing the pot in a sink*, (b) *opening the microwave*, (c) *taking off the fork*, (d) *stacking the bowls*, (e) *picking the block in pan*, and (f) *pouring water into the sink*. As shown in Fig. 7, these scenarios require the model to handle complex environments and demonstrate spatial reasoning ability.

We employ a Franka Emika Panda robotic arm with wrist and side cameras for all tasks. An additional agent-view camera is reserved for visualization only. Both teleoperated demonstrations and robot manipulations operate at a control frequency of 15 Hz. For each task, we collect 10 demonstrations, yielding 60 task-specific trajectories. For model training, we adopt a two-stage fine-tuning strategy with a batch size of 32: the model is first fine-tuned on the entire 176-demonstration dataset for 10k steps, followed by a task-specific fine-tuning on each individual task for 1k steps. Finally, we evaluate the learned policies using 10 trials per task.

Baselines. We select two representative baseline models: π_0 [51] as our 2D VLA baseline, and Spatial Forcing [12] as its 3D counterpart. Following Spatial Forcing [12], we use the π_0 as the backbone and integrate our proposed relation alignment. Our method and all baselines load the π_0 checkpoints pre-trained on the DROID dataset [57] for initialization.

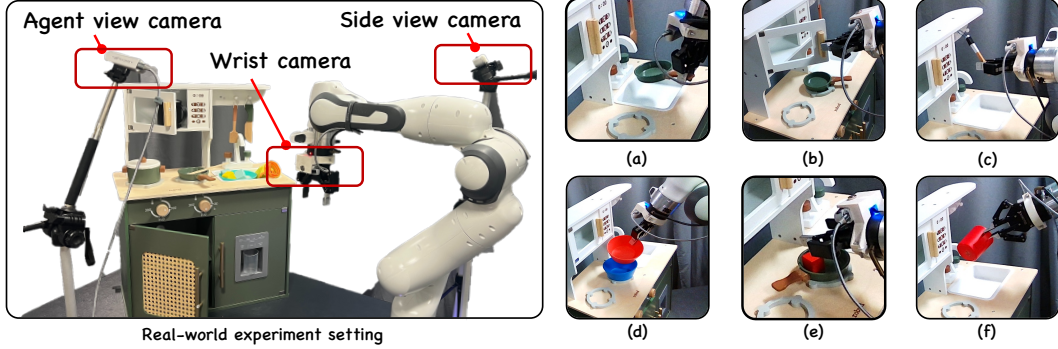


Figure 7: **Our robot environment and task examples.** We use a 7-DoF robotic arm with a two-finger gripper, a side-view camera, a wrist camera, and an agent-view camera (used only for visualization). We evaluate on six tasks in a complex kitchen environment.

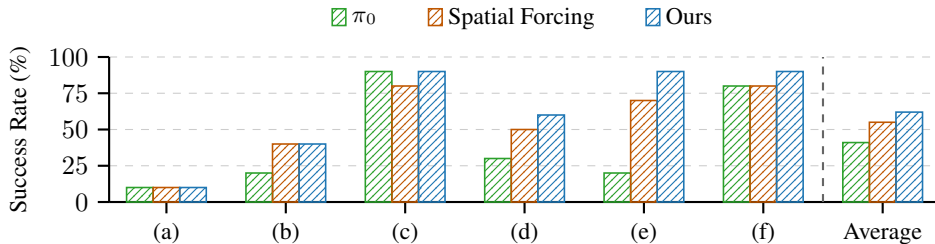


Figure 8: **Performance comparison on real-world tasks.** We report the success rates and the average rates on the six real robot tasks. All the models took 10 trials on each task. (a) – (f) are consistent with the labels described in Fig. 7 and Sec. 5.1.

5.2 Results on real-world environment

Fig. 8 shows per-task success rates across all three models. Our method achieves the best overall performance, with gains on tasks that demand precise 3D spatial reasoning. The advantage is most evident in (e) *picking the block in pan*: the policy must localize a small object from only two RGB views, and without geometric grounding, the 2D baseline π_0 fails to precisely localize the object in 3D space, while both 3D-aware models recover the correct position more reliably. For (c) *taking off the fork*, (d) *stacking the bowls*, and (f) *pouring water into the sink*, the wrist-camera view provides sufficient spatial cues for localization, so all models perform comparably well. The most challenging tasks are (a) *placing the pot in the sink* and (b) *opening the microwave*: the affordance region is small, flat, and far from the side camera, making it difficult for all the models to infer precise manipulation.

6 Discussion

Limitations. Despite its effectiveness, RIGID relies on a pre-trained 3D expert during training. Therefore, the quality of spatial distillation inherently depends on both the teacher model’s capabilities and the scale of the training data. Specifically, we use VGGT [16] as the 3D teacher; while effective and generalizable, it still possesses known biases and limitations [30, 56]. Furthermore, the limited scale of the robot fine-tuning dataset presents another constraint. Although VGGT is pre-trained on large-scale 3D data, the comparatively much smaller fine-tuning dataset restricts the diversity and generalizability of the spatial knowledge transferred to the student model.

Conclusion. We present RIGID, a framework designed to enhance spatial understanding in VLA models while remaining effective on VLA’s original capabilities. The key idea is that, unlike direct feature alignment with a 3D teacher model, RIGID distills 3D awareness into 2D VLAs while

preserving their original representation space. To effectively incorporate spatial cues, we formulate geometry distillation as a Wasserstein-based optimization problem. Experiments in simulation, real-world robotic settings, and probing analyses, together with representation rank analysis, demonstrate that our method improves spatial understanding while preserving semantic understanding.

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A Training details

Table S1 shows a default setting used in our experiments on LIBERO and real-world settings. Other settings are the same as the default setting of finetuning OpenVLA-OFT [41] or π_0 [51] in the original repositories.

Table S1: Combined hyperparameters for simulation and real-robot training configurations.

Hyperparameters	Simulation (LIBERO) Config	Real-Robot Config
Base VLA	OpenVLA (7B)	π_0 -DROID (3B)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
Learning rate	5e-4 / 5e-5 (after 20k steps)	2.5e-5 (peak)
Lr Scheduler	MultiStepLR	Cosine Decay
Warmup Steps	1,000	1,000
Decay Steps	30,000	10,000
Batch Size	8	16
Training Steps	30,000	10,000
EMA	0.999	None
distill loss coefficient	0.5	0.5
VLA align layers	24 (total 32 layers)	12 (total 18 layers)
VGGT extract layers	12	12
Fine-tuning / LoRA rank	LoRA (rank 32)	Full (no LoRA)
GPUs	NVIDIA H100 80G $\times$ 2	NVIDIA H100 80G $\times$ 1

B Additional analysis on rank collapse

Implementing details. To further investigate the impact of spatial alignment on pre-trained VLA representations, we compare the visual features of our model against three baselines: OpenVLA [41], Spatial Forcing [12], and the 3D foundation model (VGGT) [16]. For each model, we process 50 batches ($50 \times 8 = 400$ samples) of LIBERO-Spatial trajectories from the training split. We extract the hidden states from layer 24, where both our method and Spatial Forcing incorporate depth information, and retain only visual patch tokens. For VGGT, we directly use the representations from its final layer, which also serve as the distillation target in Spatial Forcing.

We then mean-pool the spatial patch tokens to obtain a single global feature vector per frame. After applying per-sample ℓ_2 normalization, we construct a feature matrix and mean-center it across samples by subtracting the feature mean from each vector before performing singular value decomposition (SVD). From the resulting singular-value spectrum, we compute three metrics to assess feature diversity: the stable rank, which reflects the number of effectively utilized dimensions; the effective rank, which reflects the uniformity of variance across dimensions; and the PCA cumulative explained variance, which indicates the degree of feature collapse into dominant dimensions.

We visualize these properties using a Scree Plot (Fig. S1(a)) and a Cumulative Variance Plot (Fig. S1(b)). While the absolute rank values may be affected by the limited sample size, our analysis focuses on the relative differences across models.

Additional analysis. In the Scree Plot (Fig. S1(a)), the log-scaled singular values across component indices reflect the representational capacity of the model, where a slowly decaying curve indicates a rich, high-dimensional feature space. As shown, the curve for our method closely aligns with that of the OpenVLA baseline. This alignment confirms that our relational spatial alignment mechanism effectively preserves the distributed, high-capacity semantic features of the original foundation model. Conversely, the curve for the Spatial Forcing baseline shows a noticeable downward shift, indicating a loss of expressive power across higher-order components. This dimensional collapse is further validated by the Cumulative Explained Variance plot (Fig. S1(b)), where a steeper ascent implies that information is heavily compressed into fewer principal components. The Spatial Forcing baseline

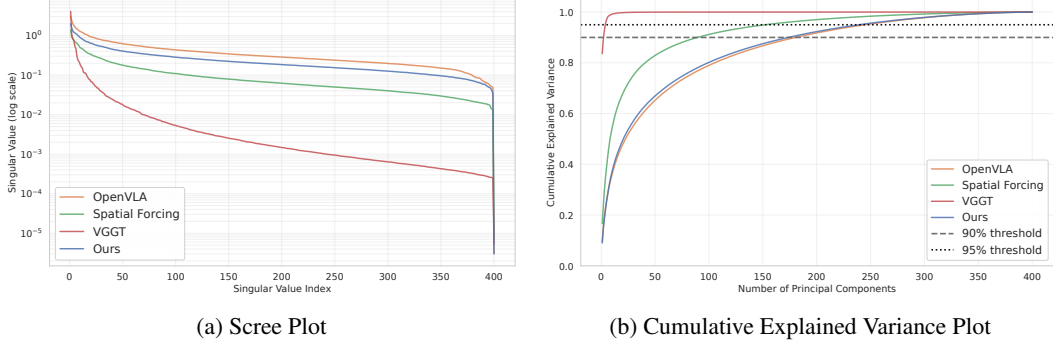


Figure S1: Dimensionality analysis of visual features.

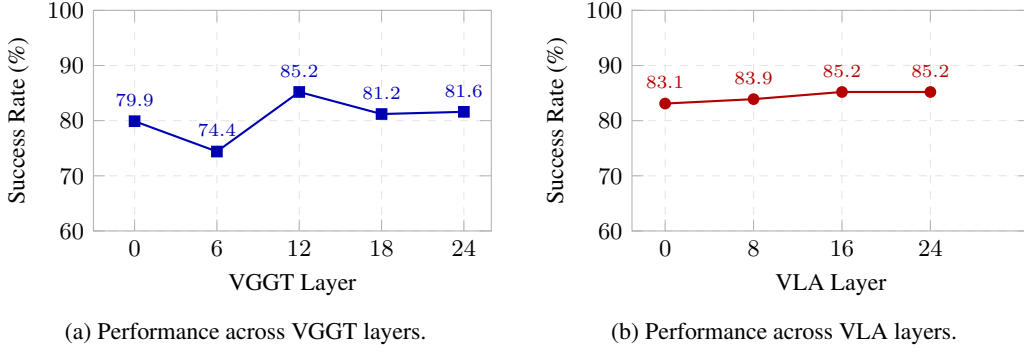


Figure S2: Success rate comparison between different layer configurations of VGGT and VLA. Layer 0 in VGGT layer means the output from the VGGT encoder. Layer 0 in the VLA layer means the output from the VLA vision encoders.

reaches the 90% variance threshold significantly earlier than the VLA baseline, demonstrating that direct spatial forcing disrupts the model’s natural feature distribution and aggressively collapses the representation into a lower-dimensional subspace.

C Ablation study

In this section, we further analyze the impact of each component in our proposed method, including different losses and other design choices. All the experiments are trained on the LIBERO training set and zero-shot tested on the LIBERO-Plus Spatial task suite with 30k training steps and 2402 evaluation episodes.

VGGT layers. To verify that the frozen geometry backbone (VGGT [16]) inherently encodes dense cross-view correspondences, we visualize its internal attention in isolation, without any VLA weights or fine-tuning. Specifically, we convert the agent and wrist views into token sequences, concatenate them, and process them through the global blocks of VGGT to compute joint attention. For a given block ℓ , we extract the post-softmax attention weights. After averaging across the attention heads, we isolate the off-diagonal block $\tilde{A}^{(\ell)} \in \mathbb{R}^{P \times P}$, which maps the query (wrist) view to the key (agent) view. For a selected seed pixel in the query view, we extract its corresponding attention row, reshape and bilinearly upsample it to the original image resolution, and apply min-max normalization. Finally, we overlay this normalized attention map onto the key view to highlight the learned correspondences. As illustrated in Fig. S3, we compare the attention maps across different layers to determine the optimal representation. We select layer 12 because it provides the most precise and localized cross-view correspondence to the query point. In contrast, shallower layers

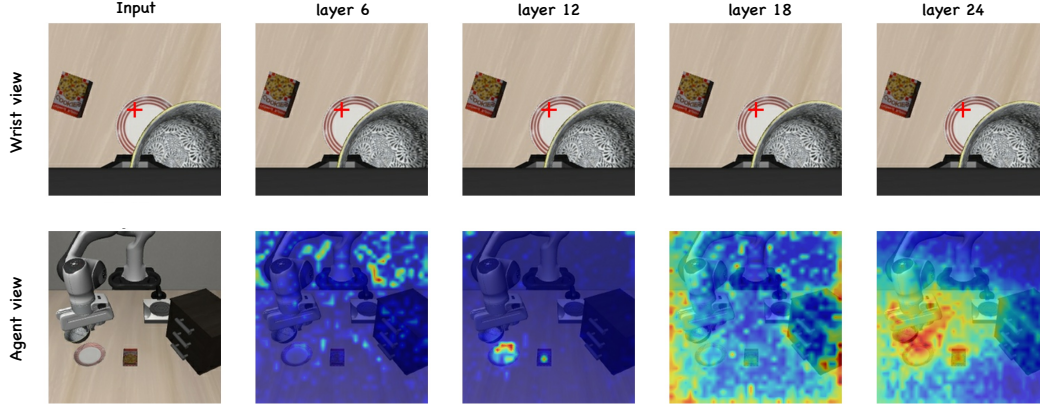


Figure S3: **Layer-wise attention comparison.**

Model	Cam	L1	L2	L3	L4	L5	Ro	Lang	Light	Back	Noise	Lay	Avg
<i>Loss</i>													
MSE	60.1	73.1	66.9	53.5	55.0	22.7	38.0	99.0	100.0	99.6	75.8	84.4	78.5
KL	73.9	83.6	76.7	69.3	70.0	59.1	52.9	88.7	97.9	98.1	88.6	95.6	84.4
L1	68.6	95.5	79.7	55.3	42.5	36.4	52.0	89.5	100.0	99.2	81.5	94.5	82.7
MMD	60.9	85.1	74.4	51.8	30.0	9.1	47.7	97.2	99.7	98.4	78.1	91.4	81.0
Wasserstein	83.2	97.0	88.0	74.6	67.5	86.4	48.1	83.6	99.7	94.2	90.3	96.1	85.2
<i>Other Ablations (based on Wasserstein loss)</i>													
(a) w/ EMA	90.4	94.0	92.5	92.9	85.0	63.6	51.1	77.4	98.6	99.2	96.0	97.9	86.6
(b) w/ distill coe.=2.0	47.1	70.1	55.6	32.5	32.5	27.3	44.0	96.2	99.7	97.3	68.1	90.6	76.4
(c) w/ attention heads	54.8	91.0	69.2	26.3	37.5	36.4	42.3	98.2	99.3	99.2	77.5	94.0	79.8
(d) w/ feature	68.6	82.1	73.7	60.5	65.0	45.5	39.3	96.9	100.0	96.9	90.0	95.6	83.9

Table S2: **Component analysis.** We systematically analyze different component choices. KEY-EMA: Exponential Moving Average, coe.: coefficient.

(e.g., layer 6) produce scattered attention distributions, while deeper layers (e.g., layers 18 and 24) yield overly diffuse patterns.

These qualitative results align well with our practical results, as shown in Fig. S2a. It also aligns well with the analysis proposed by Bratulić et al. [56], which indicates that the VGGT begins to encode the geometry information from Layer 12.

VLA layers. We also conduct an ablation study on the selection of the VLA integration layer. Consistent with Spatial Forcing [12], we empirically find that integrating spatial features at Layer 24 yields optimal performance. However, we observe only marginal performance variations across the different layers.

Loss function. We evaluated several loss functions as optimization objectives, including L1, Mean Squared Error (MSE), Kullback-Leibler (KL) divergence, Maximum Mean Discrepancy (MMD) [59, 60], and Wasserstein loss. Empirically, the Wasserstein loss achieves the best balance across all perturbation categories. While KL divergence performs well generally, it yields suboptimal results on camera view change tasks. This aligns with our previous analysis 3.2: under a KL objective, a minor spatial shift is penalized equally to a completely incorrect prediction, which provides poor gradient signals when attention distributions do not perfectly overlap. Consequently, its generalization to viewpoint variations is less stable than that of the Wasserstein loss.

For other losses, MMD utilizes a kernel-based metric measuring the distance between probability distributions and is completely position-agnostic; thus, it showed suboptimal performance on spatial understanding tasks that require precise location information. Conversely, L1 and MSE are designed to match absolute values rather than measure distribution distances. They performed poorly on spatial understanding tasks but demonstrated excellent robustness against position-invariant perturbations, such as lighting and background changes.

Considering all these factors, we select the Wasserstein loss as the primary optimization objective for our method. However, we also observe its limitations in handling language and robot state representations. Effectively combining the strengths of different loss functions remains a valuable direction for future research.

EMA. We found that the usage of Exponential Moving Average (EMA) helps stabilize the training and further improve the performance. We apply an exponential moving average over all trainable parameters (LoRA adapters + action head + alignment projectors) and evaluate with the averaged weights. We hypothesize that its benefit is attributed to two factors: the small batch size imposed by our compute budget, which makes per-step gradients noisy, and the competition among the action loss and the auxiliary alignment losses in early training; averaging the weight trajectory smooths both effects. We empirically observe that EMA consistently improves performance, as shown in our ablation study. We also perform an additional study on different EMA decay rates, as illustrated in Tab. S3. We also find that a high decay rate ($\beta = 0.9999$) yields the lowest success rate (82.1%), as overly slow EMA updates fail to track the active model, leading to evaluation on outdated parameters and reduced performance.

Table S3: Performance according to the decay parameter β on LIBERO-spatial.

EMA decay rate	0.0	0.9	0.999	0.9999
Avg. successful rate (%)	85.2	85.1	86.6	82.1

Distillation coefficient. We set the distillation coefficient to 0.5 to prevent it from overwhelming the action supervision signal. As shown in Tab. S2(b), weighing the distillation supervision more heavily than the action signal leads to suboptimal performance.

Attention heads. The original VGGT layers contain multiple attention heads. In our current design, we average the outputs across all heads. We also investigated incorporating a multi-head design into the projectors for the VLA vision features, performing distillation on a per-head basis. However, in practice, we found that this approach complicates training with a limited fine-tuning dataset and ultimately leads to suboptimal performance.

Utilization of layer feature. Our current design employs post-softmax q - k attention weights as the distillation target. We also investigate using VGGT layer features as an alternative target. As demonstrated in Tab. S2 (d), while this alternative achieves comparable or even superior performance in some categories, it performs significantly worse in the Camera category. We hypothesize this is because the q - k attention weights, having passed through the query and key projectors, more effectively filter out irrelevant information encoded within the features.

D Probing experiments

Probing dataset construction. We generate a dedicated probing dataset using the LIBERO benchmark [39], which provides 130 tabletop manipulation tasks across five task suites: LIBERO-Spatial (10 tasks), LIBERO-Object (10 tasks), LIBERO-goal (10 tasks), LIBERO-10 (10 tasks), and LIBERO-90 (90 tasks). All demonstrations are rendered in the MuJoCo physics simulator via Robosuite [61]. For each task demonstration, we render two camera views simultaneously: an agent view (fixed third-person camera) and a wrist camera. For each view we record:

- RGB images from two views.
- Depth map in meters, from the simulation depth buffer.
- Instance segmentation map of per-pixel instance IDs from the simulation scene graph. Robot body components and backgrounds are excluded and remapped to class 0. The resulting taxonomy contains 52 classes: background and 51 interactive object categories.
- Cross-View Correspondence. For each demonstration frame, we compute up to $K = 32$ valid point correspondences between agent-view and wrist-view using geometric reprojection through camera intrinsics and extrinsics provided by the simulation. Candidate query pixels are sampled uniformly from foreground pixels in the wrist-view image.

Each LIBERO task provides 50 demonstrations that often exhibit near-identical initial object configurations, introducing significant redundancy. For each task, we select $K = 5$ maximally diverse demonstrations using farthest-point sampling (FPS) on the MuJoCo state vector. FPS initializes from demo 0 and iteratively selects the demo whose state vector is farthest (in ℓ_2 distance) from all already-selected demos.

We split at the task level rather than the demo level to test generalisation to held-out tasks and object configurations. For the four main suites, we assign the 8 tasks to training and the 2 tasks to validation. The LIBERO-90 suite (90 tasks) is assigned entirely to training following the setup in the original LIBERO dataset. As a result, it contains 122 training tasks of 610 demos and 8 testing tasks with 40 demos.

Feature extraction. We extract the hidden state features from the 24th layer of OpenVLA [40], Spatial Forcing [12], and our proposed method. This specific layer is selected because the distillation loss is applied to it during training, thus directly reflecting how distillation shapes the learned representations. All probes operate on these frozen features using lightweight output heads, detailed as follows.

Depth probe. The output head architecture utilizes a 1×1 convolution to map the features to spatial scalars, followed by a Softplus activation to ensure positivity and then bilinear upsampling to input resolution. The L1 loss is applied. The evaluation metrics are $\delta_{1.25}$, $\delta_{1.03}$, and Relative Error (REL).

Segmentation probe. A 1×1 convolution projects the features into 52 class logits per spatial location, which are subsequently bilinearly upsampled. This probe is supervised via the cross-entropy loss. The evaluation metric is Pixel Accuracy.

Correspondence probe. A shared linear projection embeds both agent and wrist view features into a 256-dimensional space, generating a spatial correspondence heatmap via ℓ_2 -normalized dot-product attention. The objective is formulated as a 256-way cross-entropy classification loss applied over the flattened 16×16 token grid. The evaluation metrics are Mean Pixel Error (MPE) and PCK10.

E Real robot experiments

For demonstrations of the real-world robot experiments, we strongly encourage readers to review our supplementary videos. Additionally, Fig. S4 illustrates these experiments, detailing the six evaluation tasks and the corresponding multi-view input images. Note that the left-side view is provided solely for visualization purposes. Finally, we report typical failure modes, such as imprecise localization and physical execution errors, as shown in S5. Please refer to the supplementary video for results and failure cases.

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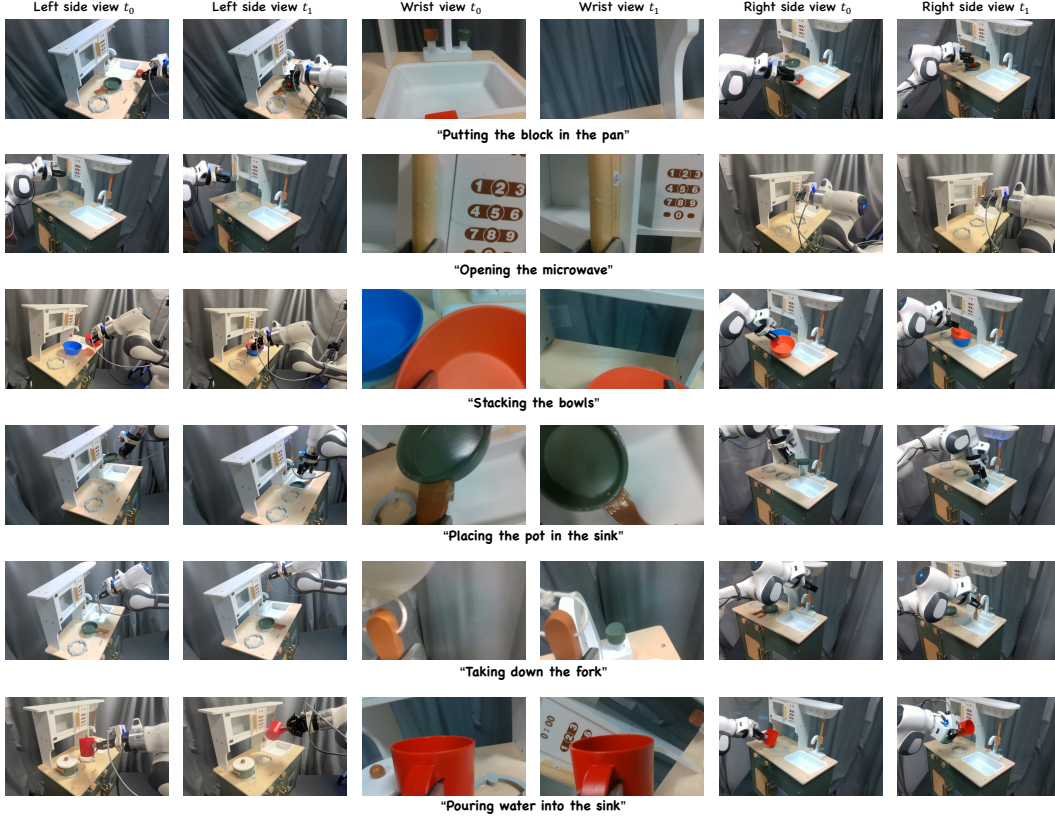


Figure S4: Real robot experiments

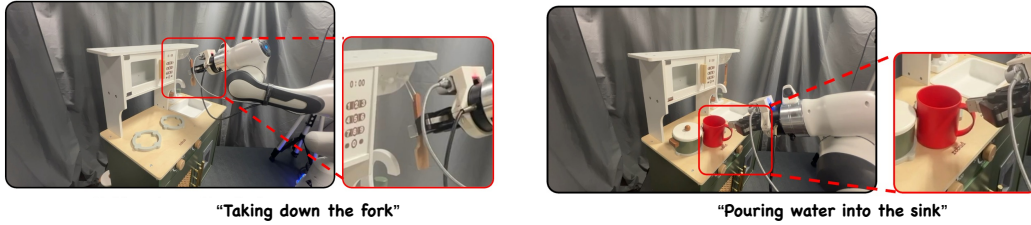


Figure S5: Failure cases

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Table S4: Statistics of demonstration counts for different tasks.

Task	Demo
block on block	9
close microwave	1
hang out spatula	10
open lid	10
open microwave	9
pan to pool	9
pick block in pan	10
pick cup in pool	9
pick out spatula	10
pick spatula in pool	9
pick spatula in the pan	8
pick up rag	10
pick up the stockpot	4
pour water	10
put cup in microwave	11
put cup out microwave	9
reach knob	15
stack bowl	10
touch 5	10
transfer boiler	3
Total (20 tasks)	176

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